JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
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An Autonomous Institute, with NAAC "A" Grade
Department of CSE\AI
"A place to learn; A chance to grow"
Session :-2025-26

Education to Eternity

MSE-1 EXAMINATION

Course: B. Tech. in CSE \ AI | **Year/Semester:** 6th Semester (Third Year)

Subject: Multimodelling Development Systems | **Subject Code:** CS6M006\AI6M006

Max Marks: 20 | **Date: 11 /02 /2026** | **Duration: - 1 Hr.**

Instructions to the Students:
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:
1. Understand the principles and practices of scalable systems for data science.
2. Apply skills for designing and implementing distributed computing systems.
3. Analyse the cloud computing technologies for data science.
4. Design and develop applications on Big Data platforms and their optimisations on commodity clusters and Clouds.
5. Implement data science algorithms and analytics using Big Data platforms.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define model-driven engineering (MDE).	L1/CO1	1
Q.2	Outline agent-based modeling.	L1/CO2	1
Q.3	Restate the term Stochastic model.	L2/CO1	1
Q.4	Describe co-simulation framework.	L2/CO2	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Restate the consistency checking and why it is important.	L2/CO2	2
Q.6	Describe the significance of modeling in engineering system design.	L2/CO1	2
Q.7	Define system dynamics and state its key elements.	L1/CO2	2
Q.8	Interpret model validation.	L2/CO1	2

SECTION C

1. **Solve any two questions.**

Q.9	Illustrate different types of models.	L3/CO1	4
Q.10	Write and explain the types of agents in agent based modelling.	L2/CO2	4
Q.11	Describe the challenges involved in multimodelling.	L2/CO1	4

All The Best